

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination May 2017

PH120, PH128 Pharmaceutical Engineering-II

Date: 15.05.17, Monday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. Which one of the following is not a mechanism of Filtration?
[A] Impact
[B] Entanglement
[C] Impingement
[D] Straining
2. Which one is not an example of filter aid?
[A] Activated charcoal
[B] Kaolin
[C] Bentonite
[D] Talc
3. Chamber press is used for
[A] Collection of precipitated antitoxin
[B] Preparation of particulate free solutions
[C] Clarification of syrups
[D] All of the above
4. Molecular distillation is also known as _____.
[A] Short Path distillation
[B] Centrifugal distillation
[C] Long Path distillation
[D] Rapid chain distillation
5. In fractionating distillation column, gasoline is obtained at _____ °C
[A] 20 - 200
[B] 10 - 100
[C] 40 - 200
[D] Less than 50
6. Who proposed that the capillaries found in the filter are highly irregular and non-uniform?
[A] Darcy
[B] Kozeny- Carman
[C] Stokes
[D] Poiseuille
7. Fluidised Bed Dryer has minimum capacity of _____ for drying.

- [A] 2 kg
 - [B] 5 kg
 - [C] 0.5 kg
 - [D] 10 kg
8. Humidifier is used for _____.
 - [A] Producing moisture level
 - [B] Removing moisture
 - [C] Decreasing moisture
 - [D] None of the above
 9. Centrifugal effect counter acts one of the following process.
 - [A] Brownian Forces
 - [B] Gravitational forces
 - [C] Cohesive forces
 - [D] Electrostatic forces
 10. Calandria consists of a number of _____.
 - [A] baffles
 - [B] outlets
 - [C] jackets
 - [D] tubular surfaces
 11. Sublimation is used as a principle in _____ dryer.
 - [A] Fluidised Bed Dryer
 - [B] Vacuum
 - [C] Freeze
 - [D] Spray
 12. Corrosion-erosion is a disadvantage of
 - [A] Evaporating pan
 - [B] Evaporating still
 - [C] Forced film evaporator
 - [D] Multiple effect evaporator
 13. In spray dryer, the bottom of the dryer is connected to a _____.
 - [A] Cyclone separator
 - [B] Atomiser
 - [C] Product collector
 - [D] Vapour controller
 14. In _____ the drying is done at below room temperature.
 - [A] Freeze dryer
 - [B] Vacuum dryer
 - [C] Tray dryer
 - [D] Fluidised Bed Dryer
 15. The materials having the tendency to absorb the moisture and to get liquified is known as _____.
 - [A] Tough
 - [B] Hygroscopic
 - [C] Efflorescent
 - [D] Deliquescent
 16. Lab scale vacuum dryer provides surface area in range of _____ metre square.
 - [A] 30-50
 - [B] 45-50
 - [C] 70-80
 - [D] 10-50
 17. Recovery of carotenoids from palm oil - is the application for _____.
 - [A] Flash

- [B] Molecular
[C] Steam
[D] Fractional
- 18 Humid volume is the volume occupied by a unit mass of dry _____ and its associated vapour.
[A] gas
[B] gas + liquid
[C] liquid
[D] none of the above
- 19 _____ is used as absorbent agent for removing moisture.
[A] Wax
[B] Gelatin
[C] MCC
[D] Silica gel
- 20 Triglycerides are distilled at _____°C onwards
[A] 150
[B] 50
[C] 100
[D] 200

SECTION – II

- Q 2 Attempt any TWO of the following**
- A Write a short note on filter leaf. 05
B Write a short note on Cartridge filter. 05
C Write a short note on Edge filter. 05
- Q 3 Attempt any TWO of the following**
- A Explain in brief about the working of sedimentation type centrifuge. 05
B Give application of basket centrifuge with proper example. 05
C Write a short note on Duhring's Rule. 05

SECTION – III

- Q 4 Attempt any FOUR of the following**
- A Write about multiple effect evaporators in brief. 05
B Which process is effective – evaporation and drying? How? 05
C Explain in brief about Vapor liquid equilibrium relationship. 05
D Differentiate between batch wise and continuous distillation. 05
E Write a short note on columns used in distillation with their applications. 05
F Define rectification. Mention the steps of it in brief. 05
- Q 5 Attempt any FOUR of the following**
- A Explain in brief about the working of vacuum dryer. 05
B Write the short note on freeze dryer and its applications. 05
C Explain in brief about the working and industrial applications of Fluidised Bed Dryer. 05
D Explain how the following terms can be obtained through Humidity chart. (Using temperature from 40°C to 32°C) 05
- Wet bulb temperature
- Relative humidity
E Draw a neat clean diagram of A.C. with its working. 05
F Explain the difference between humidification and dehumidification by the mechanism. 05